gpeyt1-38 · Sliding FS with passive

GEO PE · Long Calculation Problems I — Topic 8: Retaining Structures

Problem

A cantilever wall has total vertical load (concrete + soil on heel) ΣV = 15,400 lb/ft and horizontal active thrust Ph = 6,200 lb/ft. The base is 9 ft wide and cast on sand with base friction δb = 26° (ca = 0). Soil in front of the toe is 3 ft deep (γ = 120 pcf, φ = 30°) and the designer will count full Rankine passive resistance on the toe. Overturning FS was already checked at 2.4.

Most nearly, the factor of safety against sliding is:

A. 1.21
B. 1.47
C. 1.24
D. 1.73

Given equations / data

> FSs = [(ΣV) tan δb + ca B + Pp]/Ph; Kp = tan²(45 + φ/2); Pp = ½Kpγ Df²

Worked solution

1. Base friction = 15,400 × tan26 = 7,511 lb/ft
2. Kp = 3.00; Pp = ½(3.00)(120)(3)² = 1,620 lb/ft
3. FSs = (7,511 + 1,620)/6,200 = 1.47

Final answer

1.47

Why the other values are wrong

- **1.21** — ignored passive resistance
- **1.24** — used Ka instead of Kp in front
- **1.73** — forgot the ½ in Pp

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 Factor of Safety against Sliding (cantilever/counterfort wall figure), p. 209; §3.1.2 Rankine

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=90zXmI_yn_g, https://www.youtube.com/watch?v=F2golrHWDO4 (no creator values copied).

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